

Cytology

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Cytogenetics

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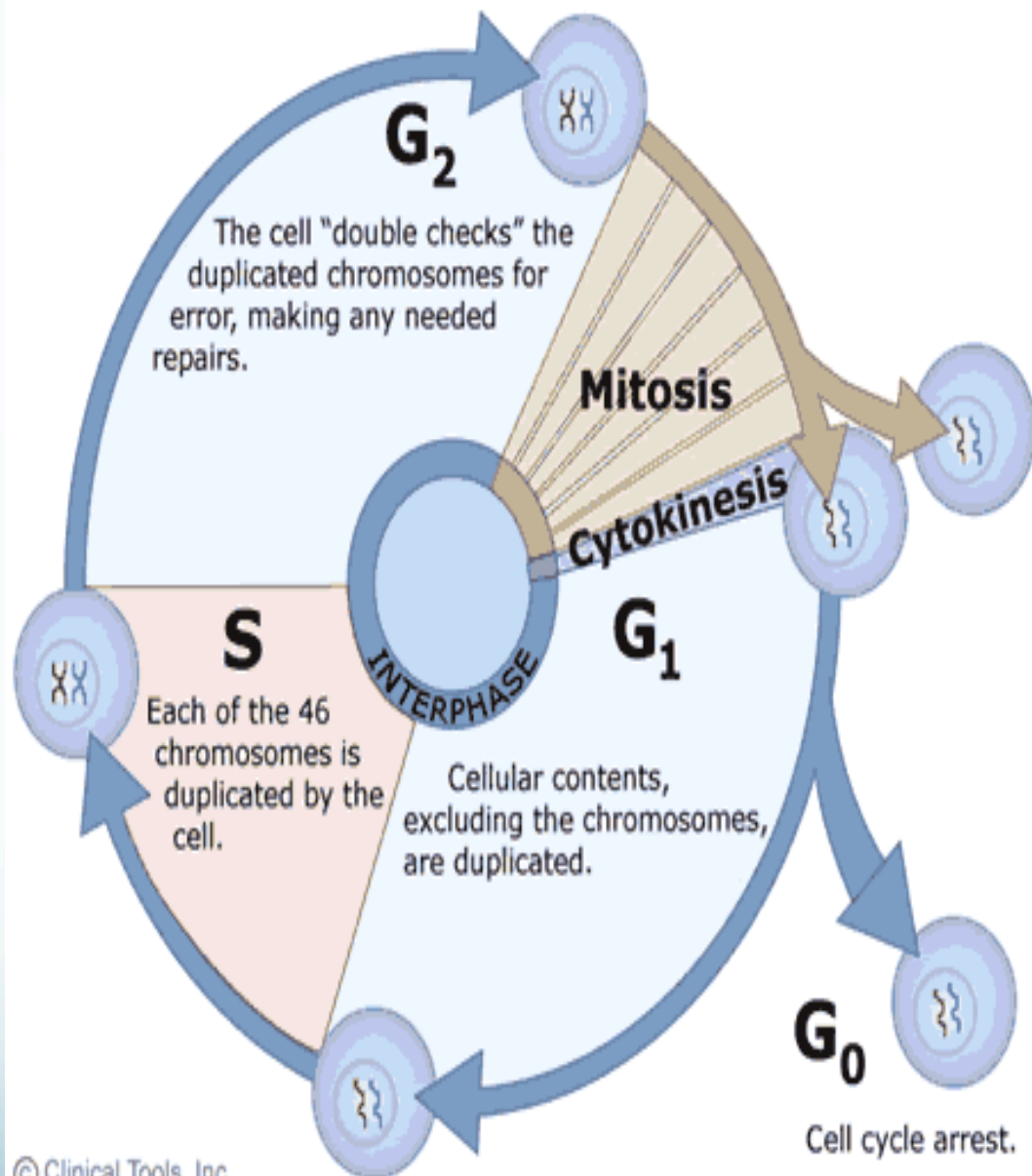
College of Medicine

Microbiology dep.

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Lecture.5

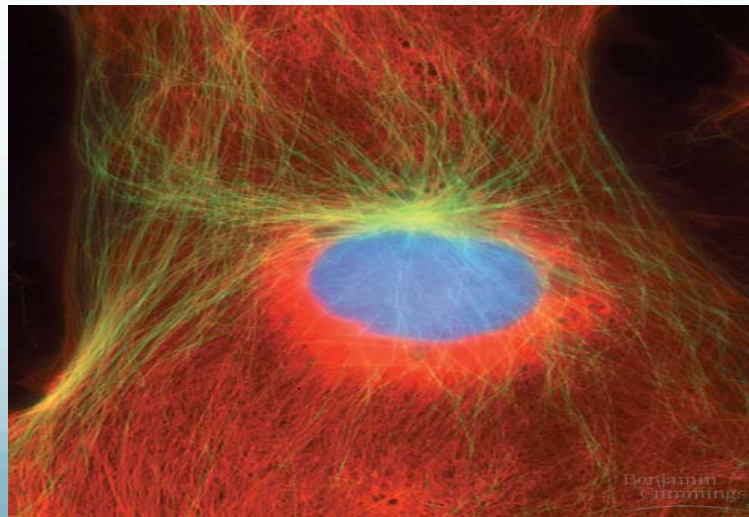
Cell division(mitosis & meiosis)



Mitosis

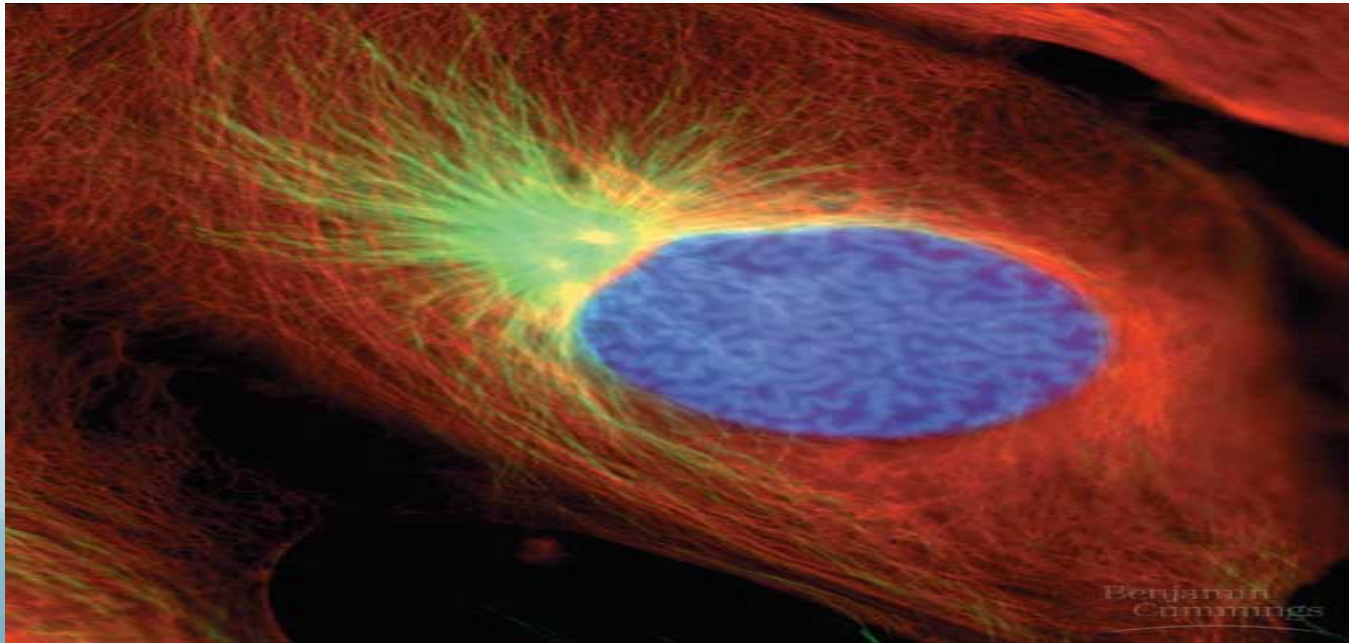
- For description, mitosis is usually broken into five sub phases:
 - 1-prophase,
prometaphase
 - 2-metaphase,
 - 3-anaphase, and
 - 4-telophase.

- By late interphase, the chromosomes have been duplicated but are loosely packed.
- The centrosomes have been duplicated and begin to organize microtubules into an aster (“star”).



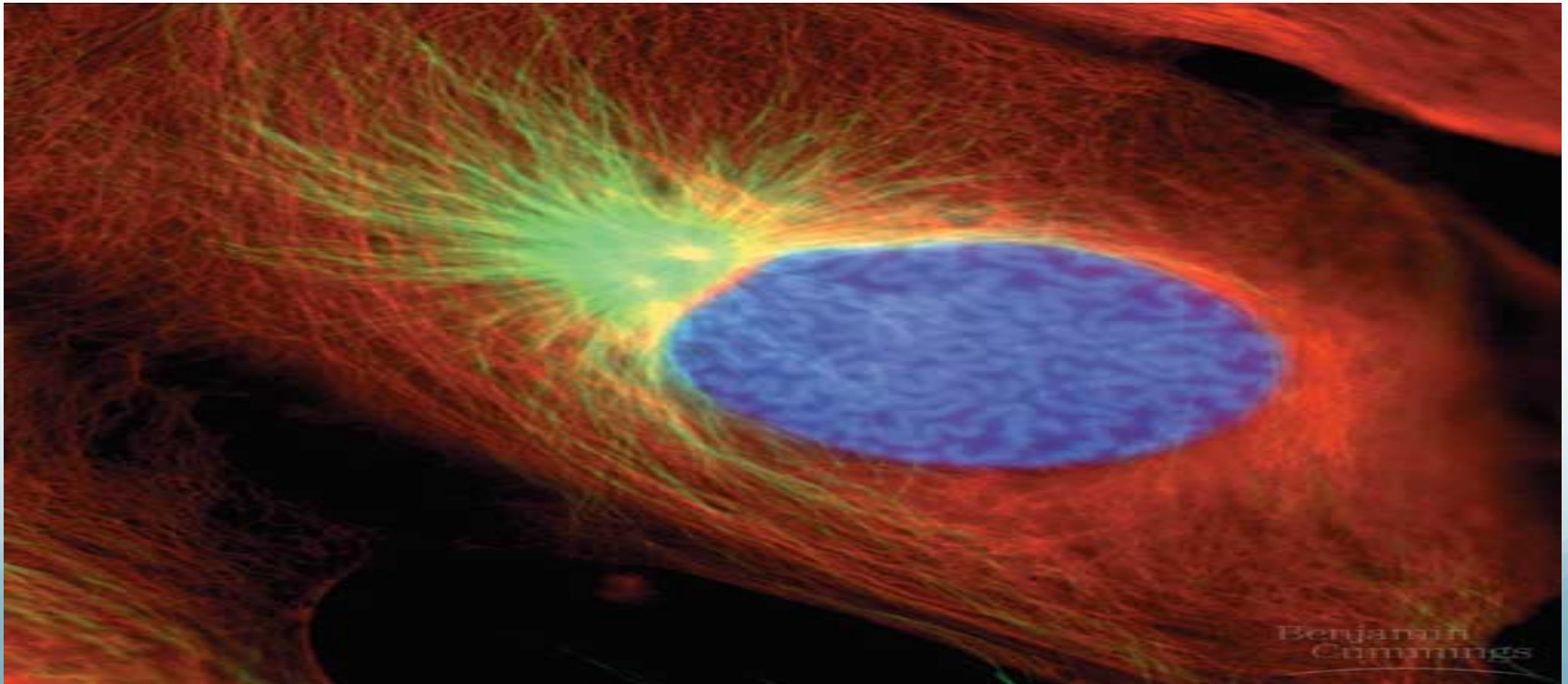
Prophase

- In prophase, the chromosomes are tightly coiled, with sister chromatids joined together.
- The nucleoli disappear.



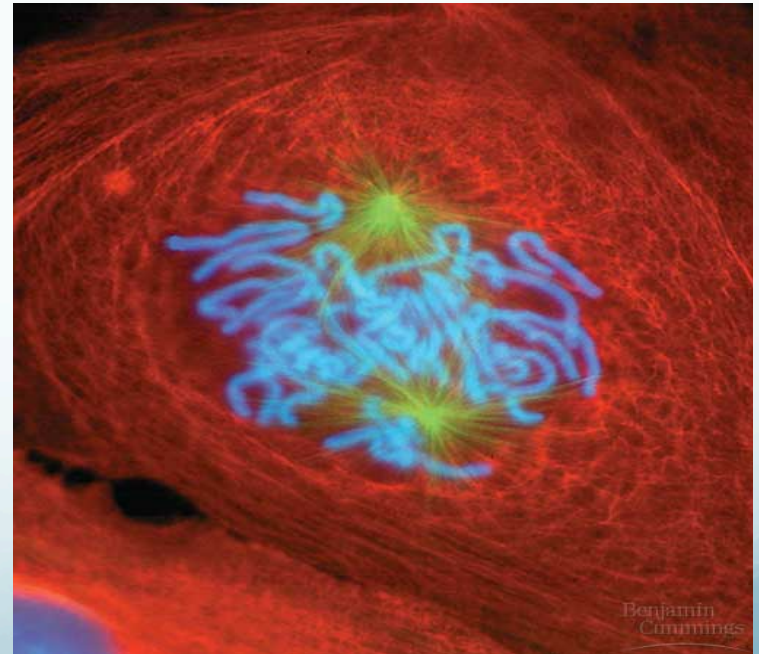
Prophase

- The mitotic spindle begins to form and appears to push the centrosomes away from each other toward opposite ends (poles) of the cell.



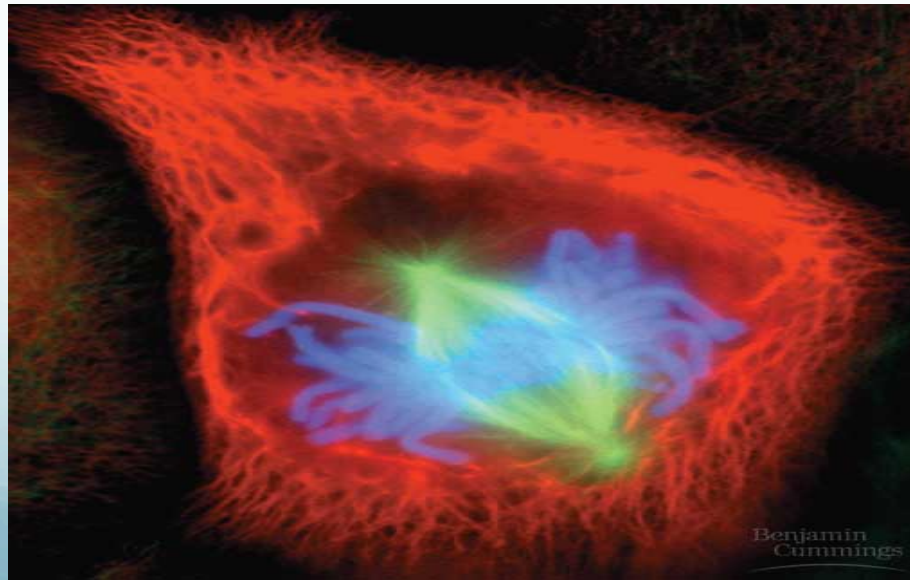
Prometaphase

- During prometaphase, the nuclear envelope fragments and microtubules from the spindle interact with the chromosomes.
- Microtubules from one pole attach to one of two **kinetochores**, special regions of the centromere, while microtubules from the other pole attach to the other kinetochore.



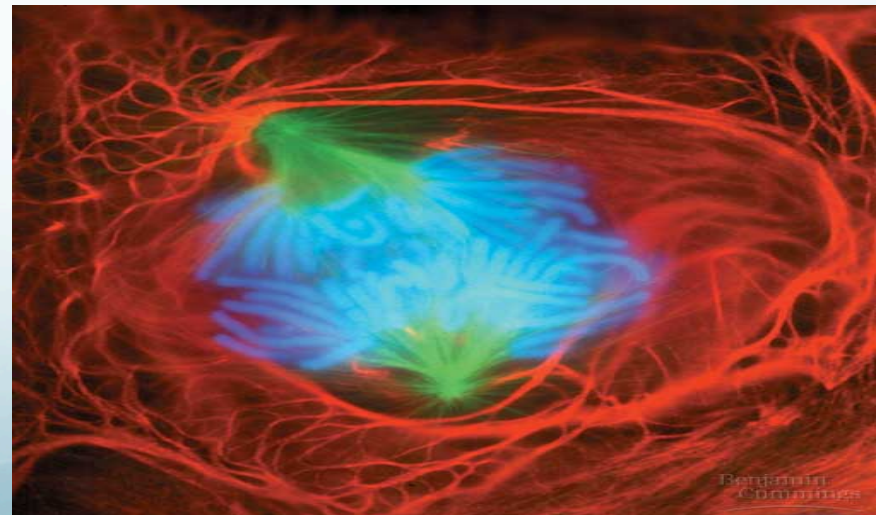
Metaphase

- The spindle fibers push the sister chromatids until they are all arranged at the **metaphase plate**, an imaginary plane equidistant between the poles, defining metaphase.



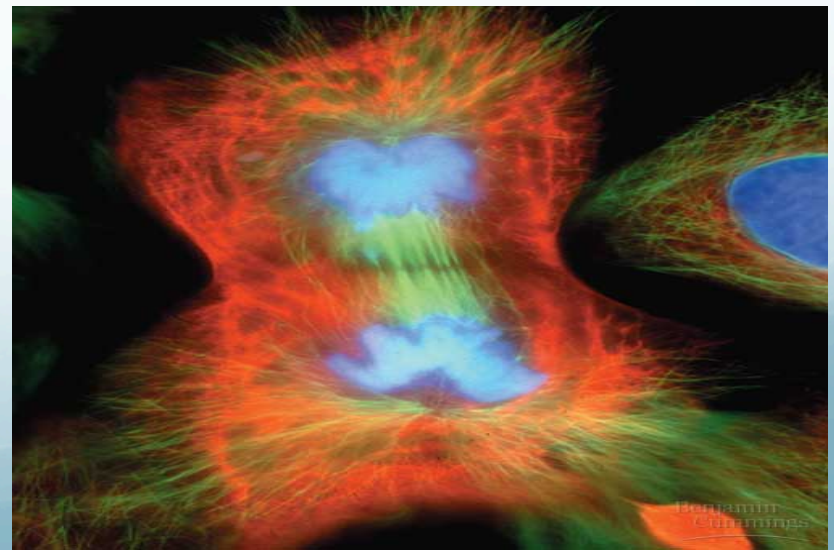
Anaphase

- At anaphase, the centromeres divide, separating the sister chromatids.
- Each is now pulled toward the pole to which it is attached by spindle fibers.
- By the end, the two poles have equivalent collections of chromosomes.



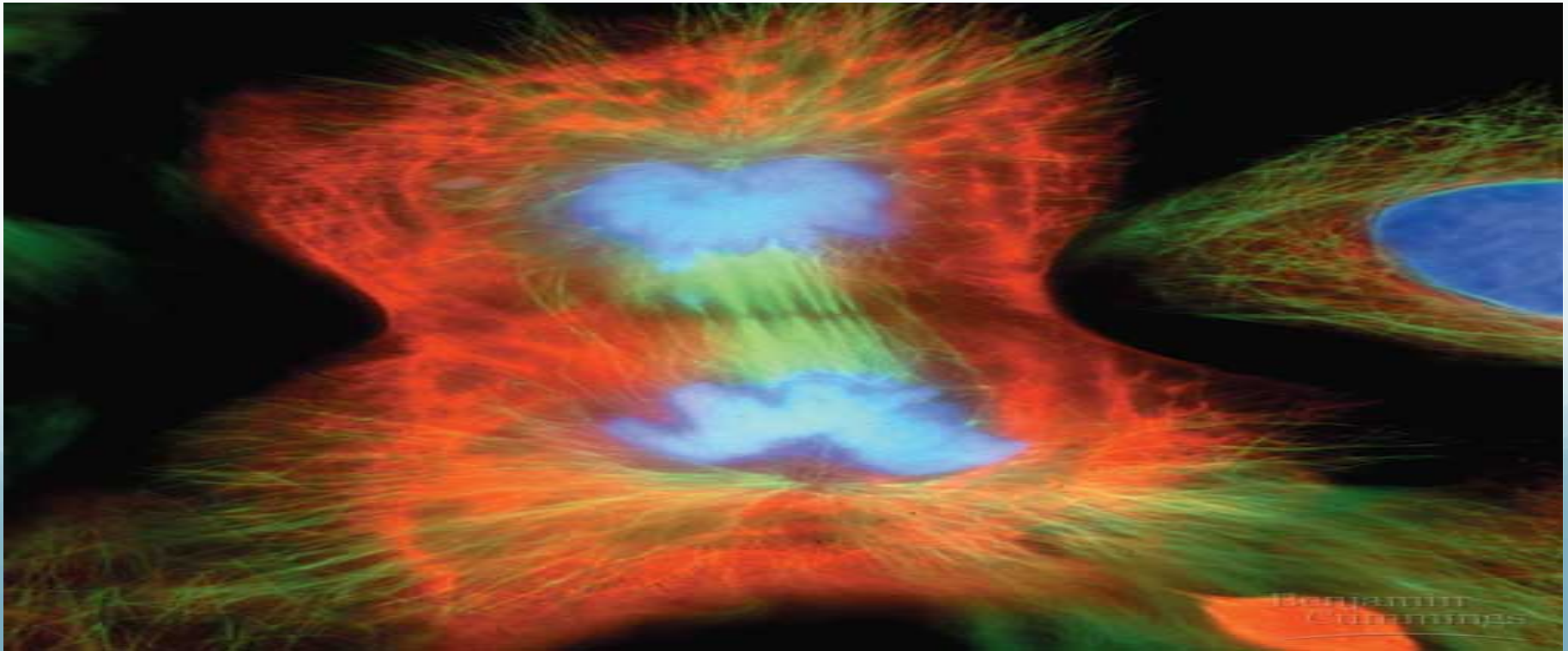
Telophase

- At telophase, the cell continues to elongate as free spindle fibers from each centrosome push off each other.
- Two nuclei begin to form, surrounded by the fragments of the parent's nuclear envelope.
- Chromatin becomes less tightly coiled.



Cytokinesis,

- Cytokinesis, division of the cytoplasm, begins after formation of two nuclei and two new cells formed, which resemble to mother cell and contain $2n$ chromosome.



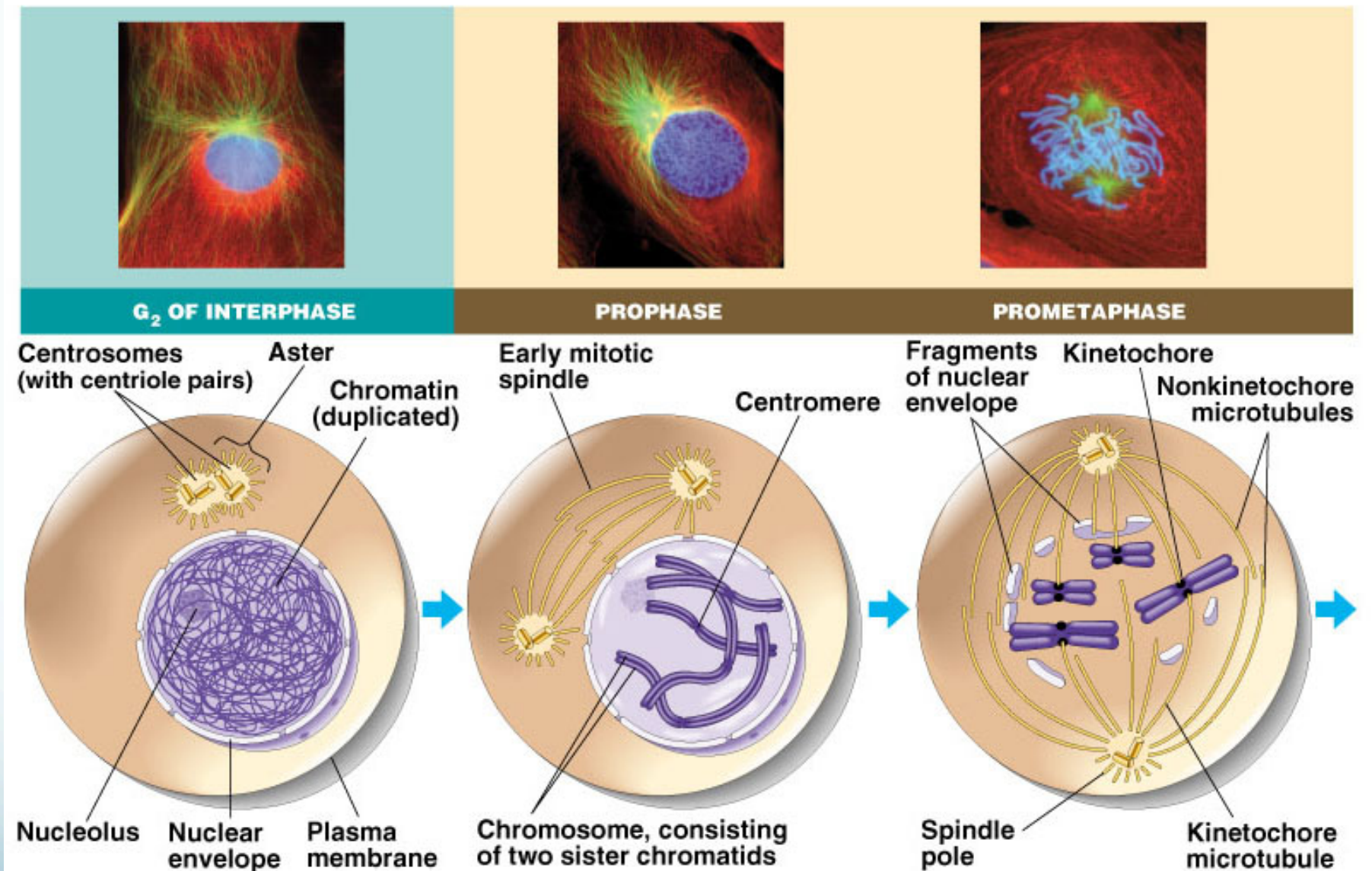
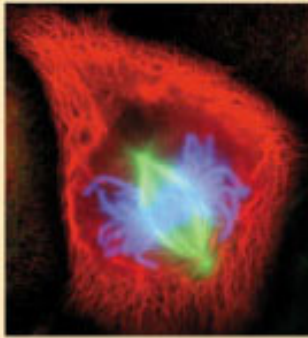
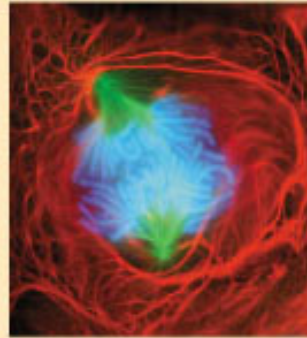


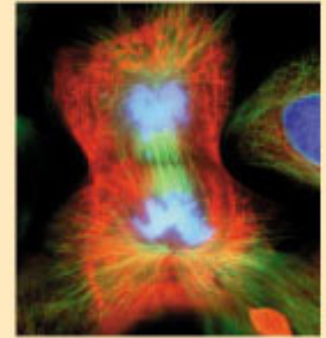
Fig. 2



METAPHASE



ANAPHASE



25 μ m

TELOPHASE AND CYTOKINESIS

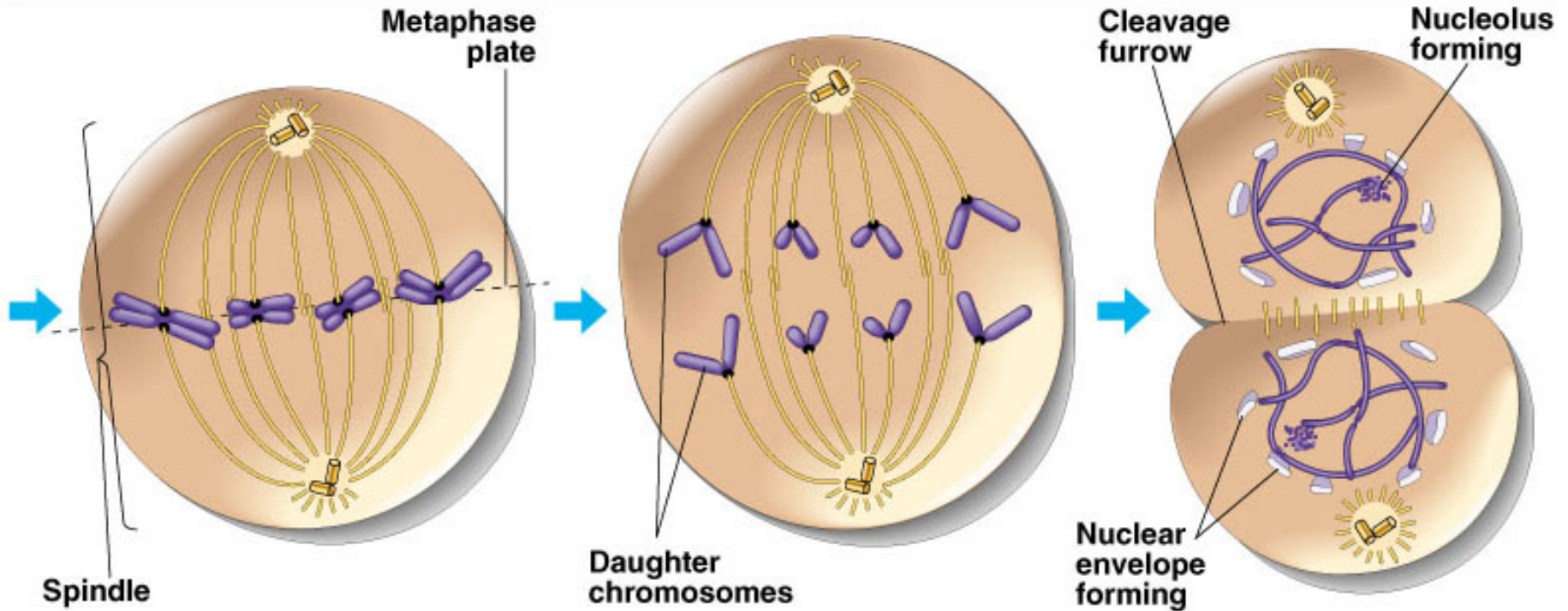


Fig. 2.

- Meiosis
- MeiosisI
 - Prophase I
 - Prometaphase I
 - Metaphase I
 - Anaphase I
 - Telophase I
 - Cytokinesis
- MeiosisII

Meiosis

- Meiosis is the form of eukaryotic cell division that produces haploid sex cells or gametes (which contain a single copy of each chromosome) from diploid (which contain two copies of each chromosome).

Meiosis

- The process takes the form of one DNA replication followed by two successive nuclear and cellular divisions (Meiosis I and Meiosis II).
- As in mitosis, meiosis is preceded by a process of DNA replication that converts each chromosome into two sister chromatids.

Meiosis I

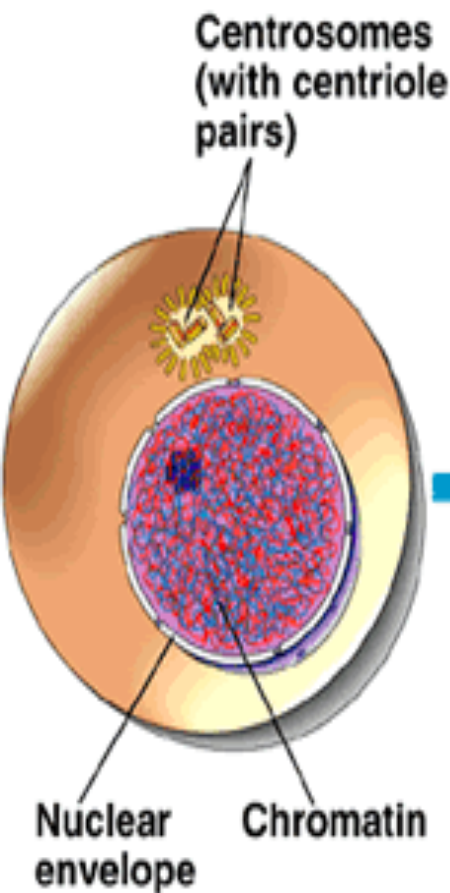
- Meiosis I
- Meiosis I separates the pairs of homologous chromosome In Meiosis I a special cell division reduces the cell from diploid to haploid.

Meiosis I

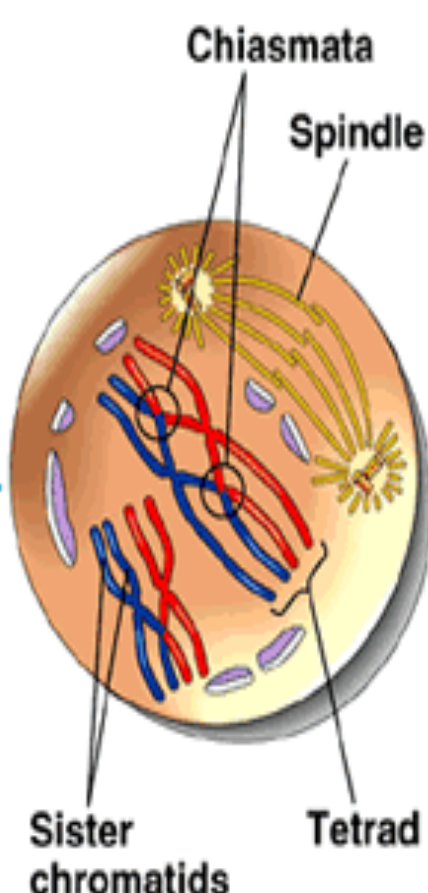
- Meiosis I is usually broken into five sub phases:
 - Prophase I
 - Prometaphase I
 - Metaphase I
 - Anaphase I
 - Telophase I
 - Cytokinesis

Meiosis I

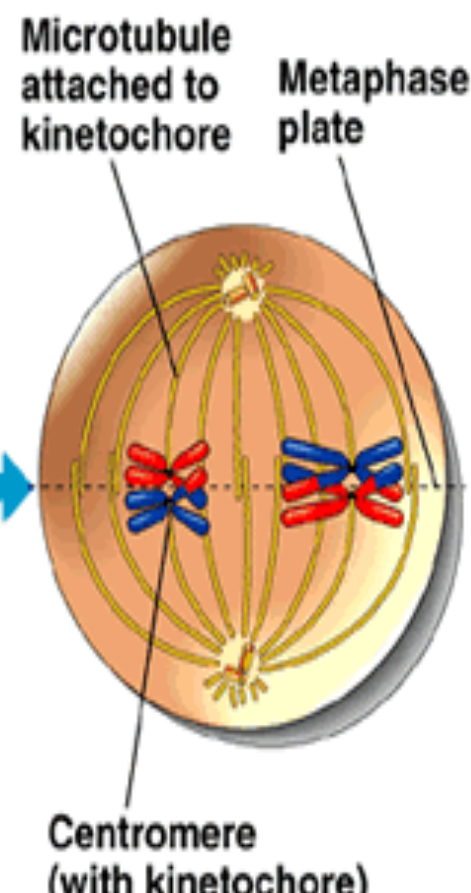
interphase I



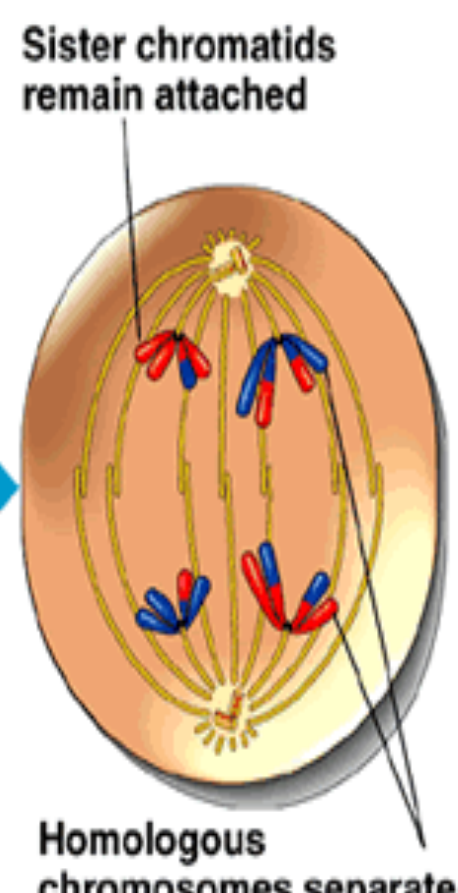
prophase I



metaphase I

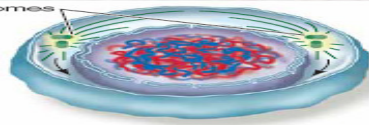
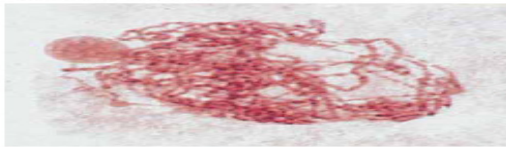


anaphase I



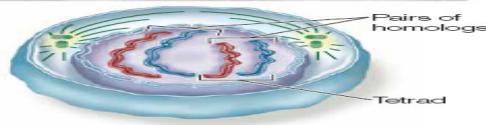
MEIOSIS I

Early prophase I



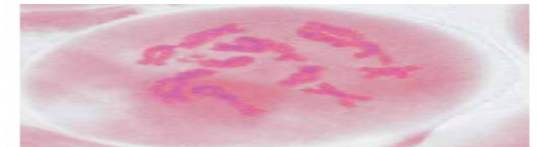
1 The chromatin begins to condense following interphase.

Mid-prophase I



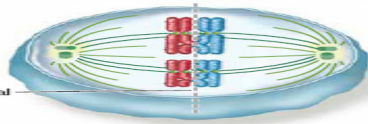
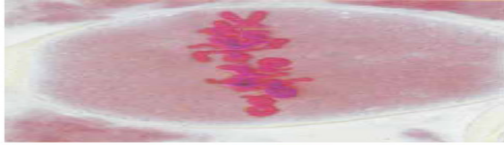
2 Synapsis aligns homologs, and chromosomes condense further.

Late prophase I—prometaphase



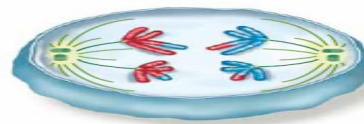
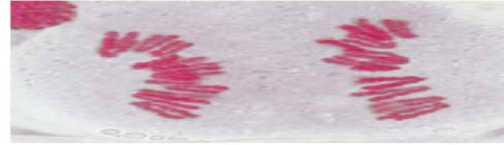
3 The chromosomes continue to coil and shorten. Crossing over results in an exchange of genetic material. In prometaphase the nuclear envelope breaks down.

Metaphase I



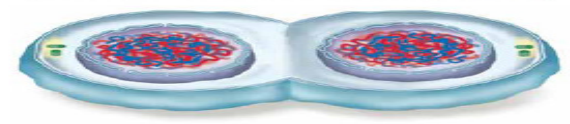
4 The homologous pairs line up on the equatorial (metaphase) plate.

Anaphase I



5 The homologous chromosomes (each with two chromatids) move to opposite poles of the cell.

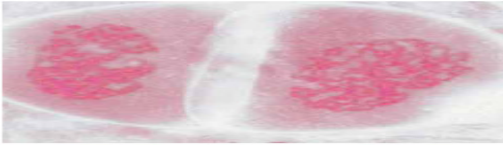
Telophase I



6 The chromosomes gather into nuclei, and the original cell divides.

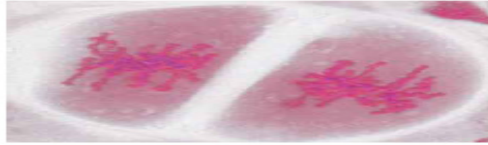
MEIOSIS II

Prophase II



7 The chromosomes condense again, following a brief interphase (interkinesis) in which DNA does not replicate.

Metaphase II



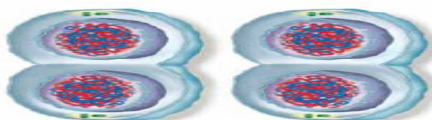
8 The centrosomes of the paired chromatids line up at the equatorial plates of each cell.

Anaphase II



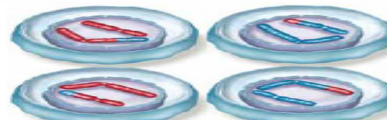
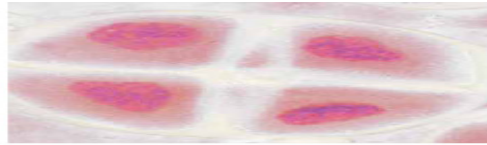
9 The chromatids finally separate, becoming chromosomes in their own right, and are pulled to opposite poles. Because of crossing over in prophase I, each new cell will have a different genetic makeup.

Telophase II



10 The chromosomes gather into nuclei, and the cells divide.

Products

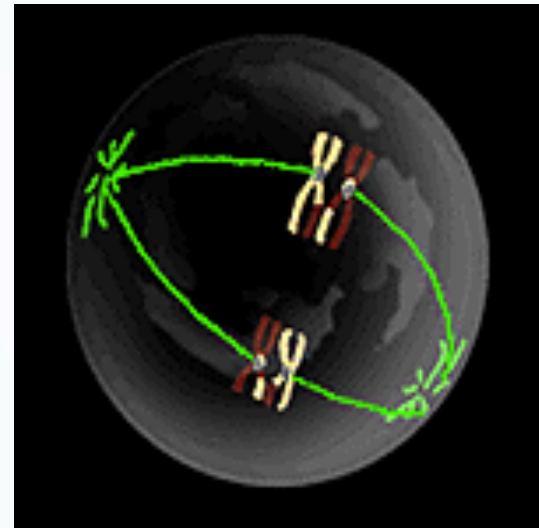
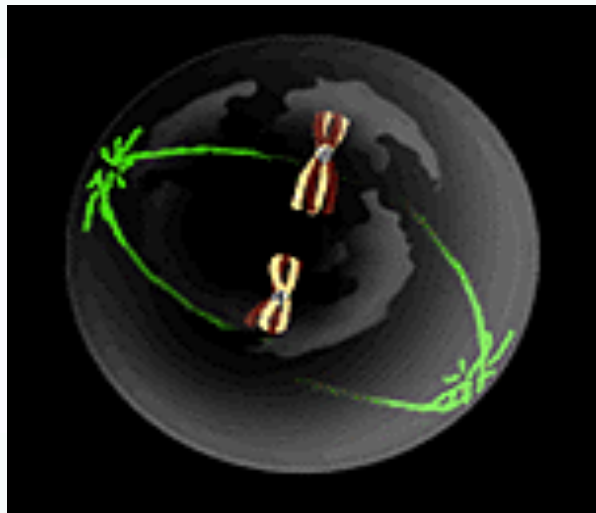
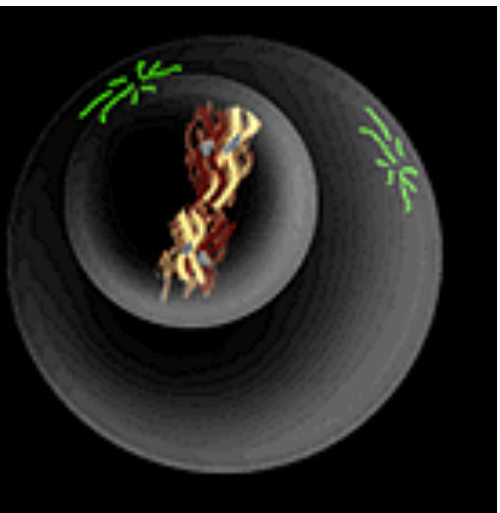


11 Each of the four cells has a nucleus with a haploid number of chromosomes.

● Prophase I

Prometaphasel

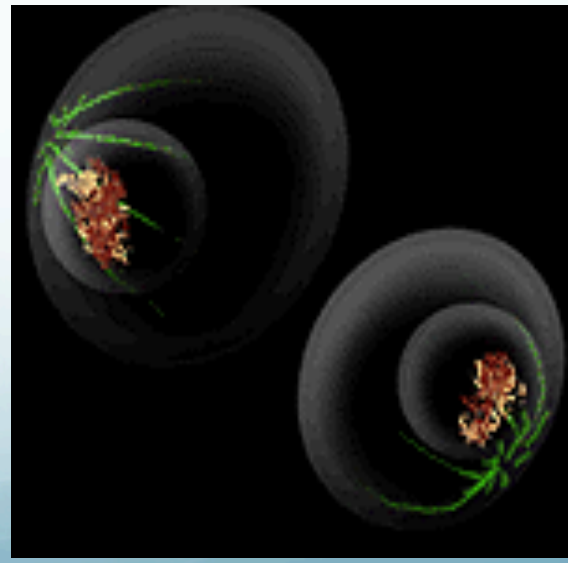
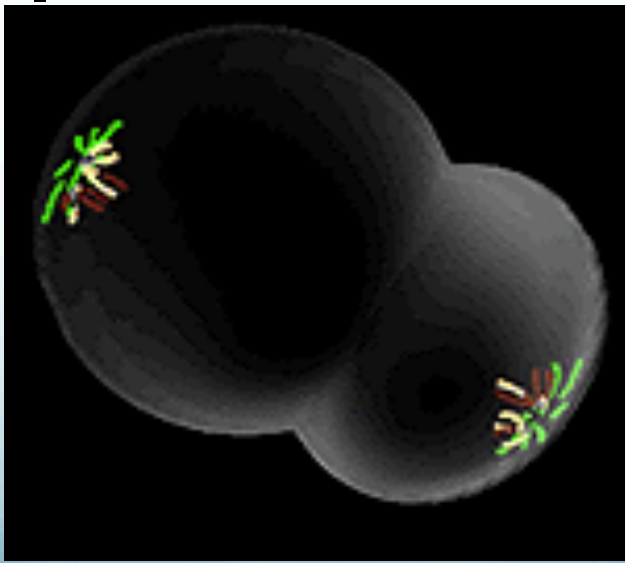
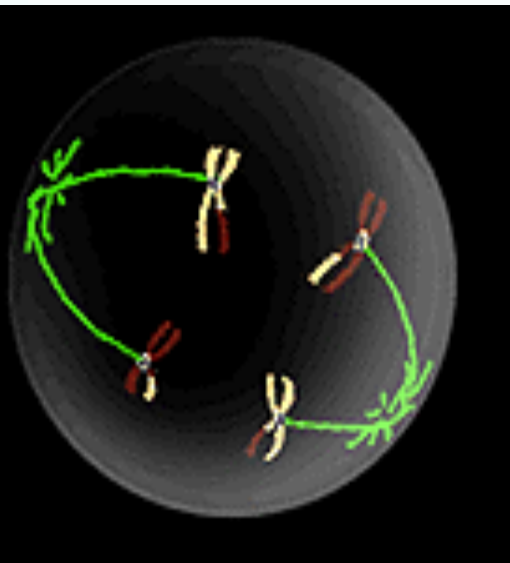
Metaphase I



Anaphase I

Telophase I

Cytokinesis is

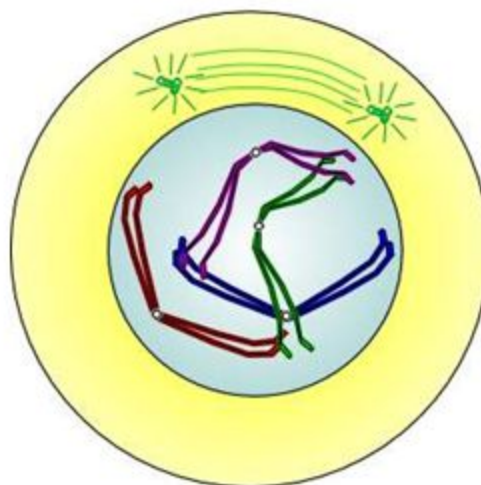


Prophase

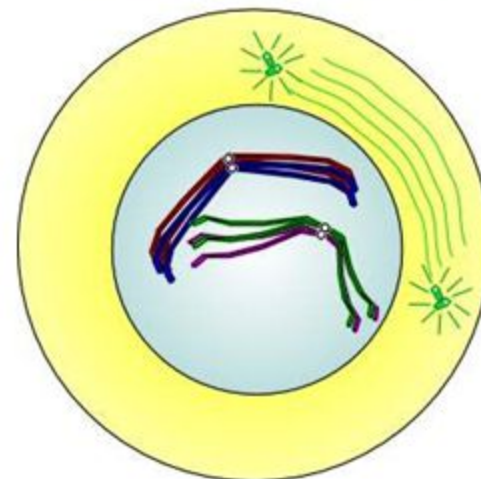
- The homologous chromosomes pair and exchange DNA to form recombinant chromosomes.
- Prophase I is divided into five phases:
- -Leptotene.
- -Zygotene.
- -Pachytene.
- -Diplotene.
- -Diakinesis.

Prophase I

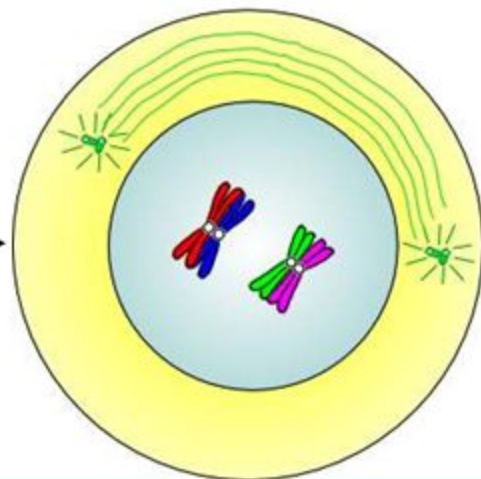
1. Leptotene



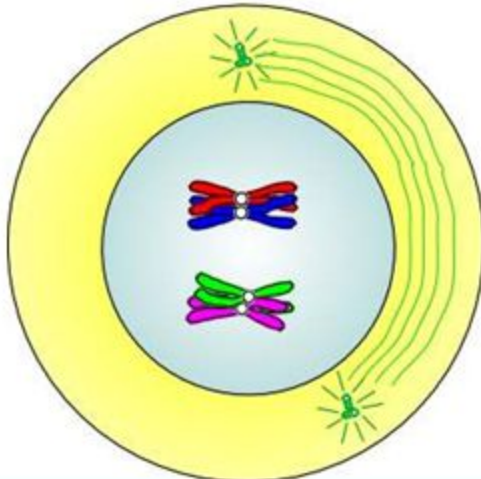
2. Zygotene



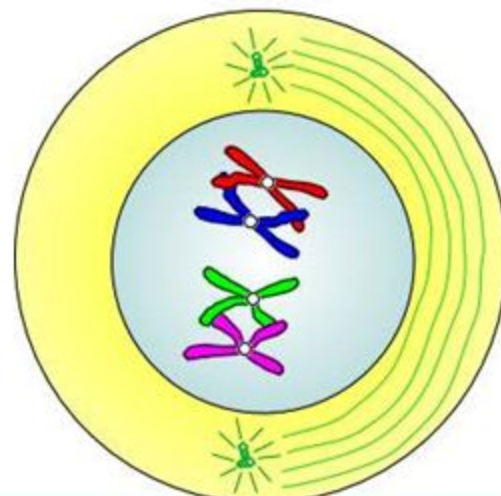
3. Pachytene



4. Diplotene



5. Diakinesis



Prophase I

- **Leptotene:**
 - chromosomes start to condense.
- **Zygotene:**
 - homologous chromosomes become closely associated (synapsis) to form pairs of chromosomes (bivalents) consisting of four chromatids (tetrads).

Prophase I

- **Pachytene:**

crossing over between pairs of homologous chromosomes to form chiasmata (sing. Chiasma)

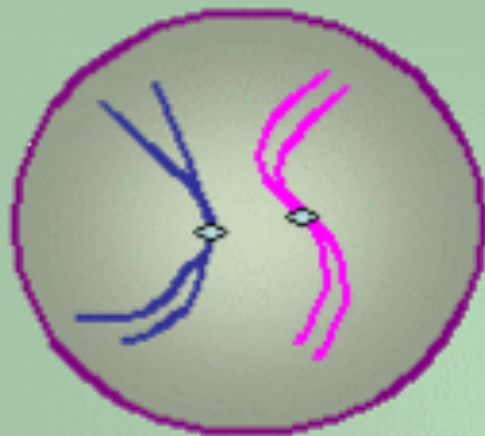
Diplotene:

homologous chromosomes start to separate but remain attached by chiasmata

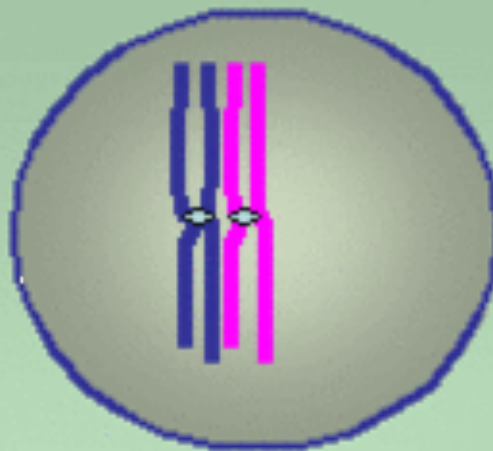
- **Diakinesis:**

- Chromosomes, homologous chromosomes continue to separate, and chiasmata move to the ends of the chromosomes.

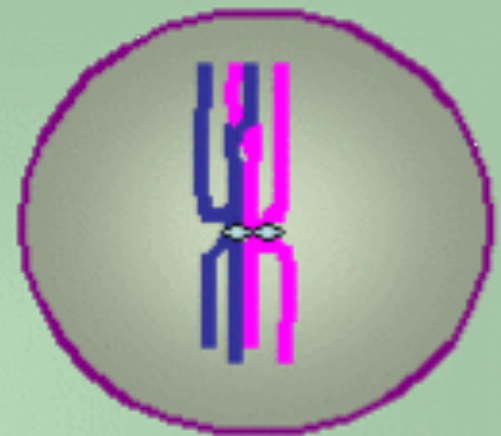
MEIOSIS (Prophase – I)



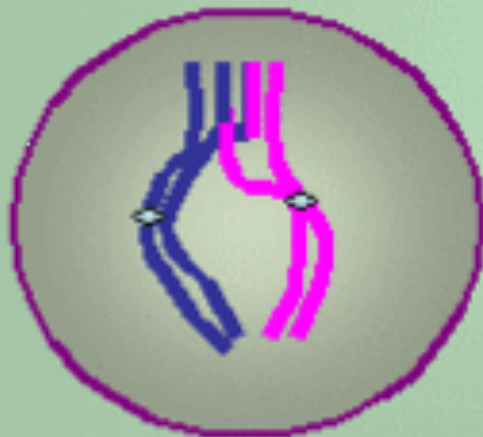
LEPTOTENE



ZYGOTENE



PACHYTENE

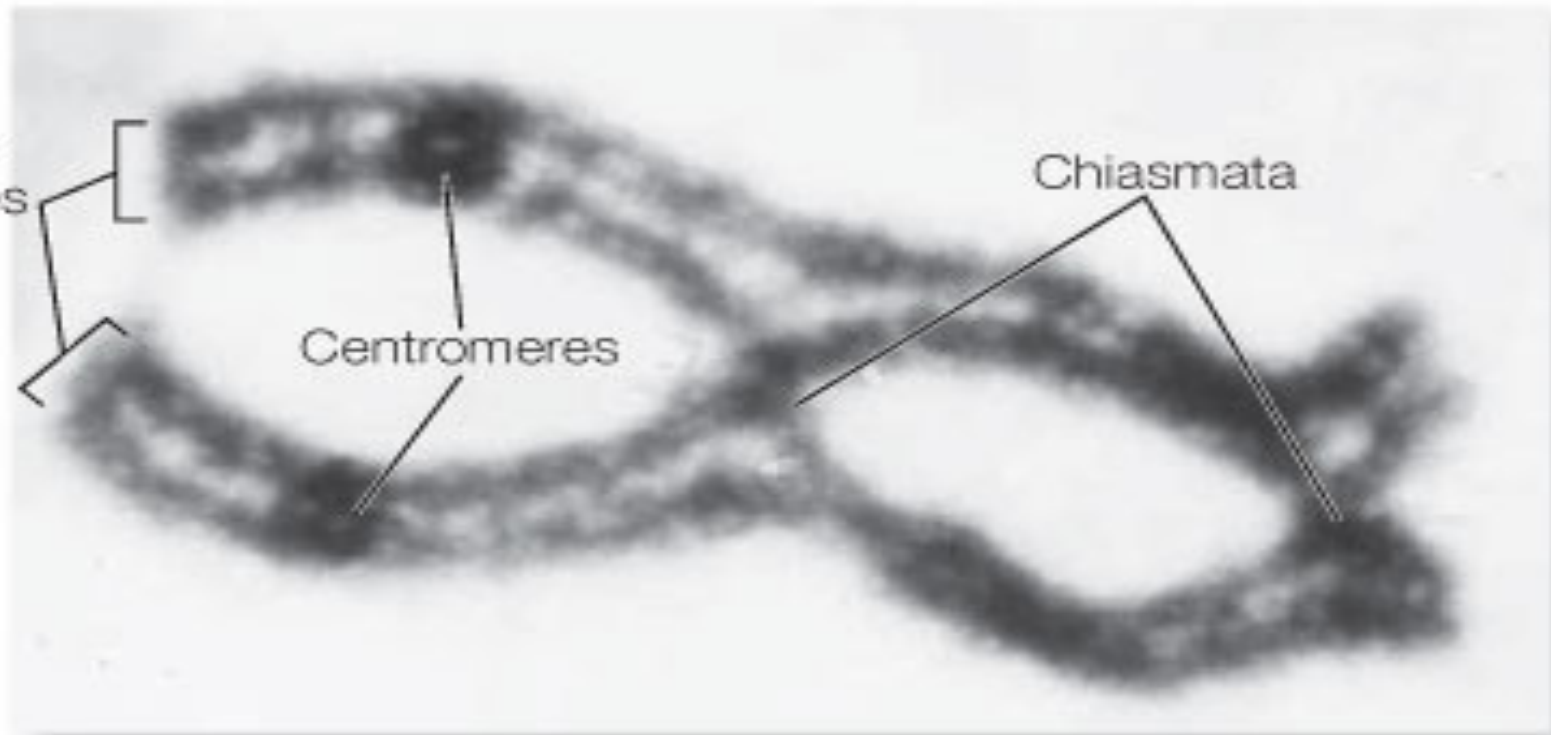


DIPLOTENE



DIAKINESIS

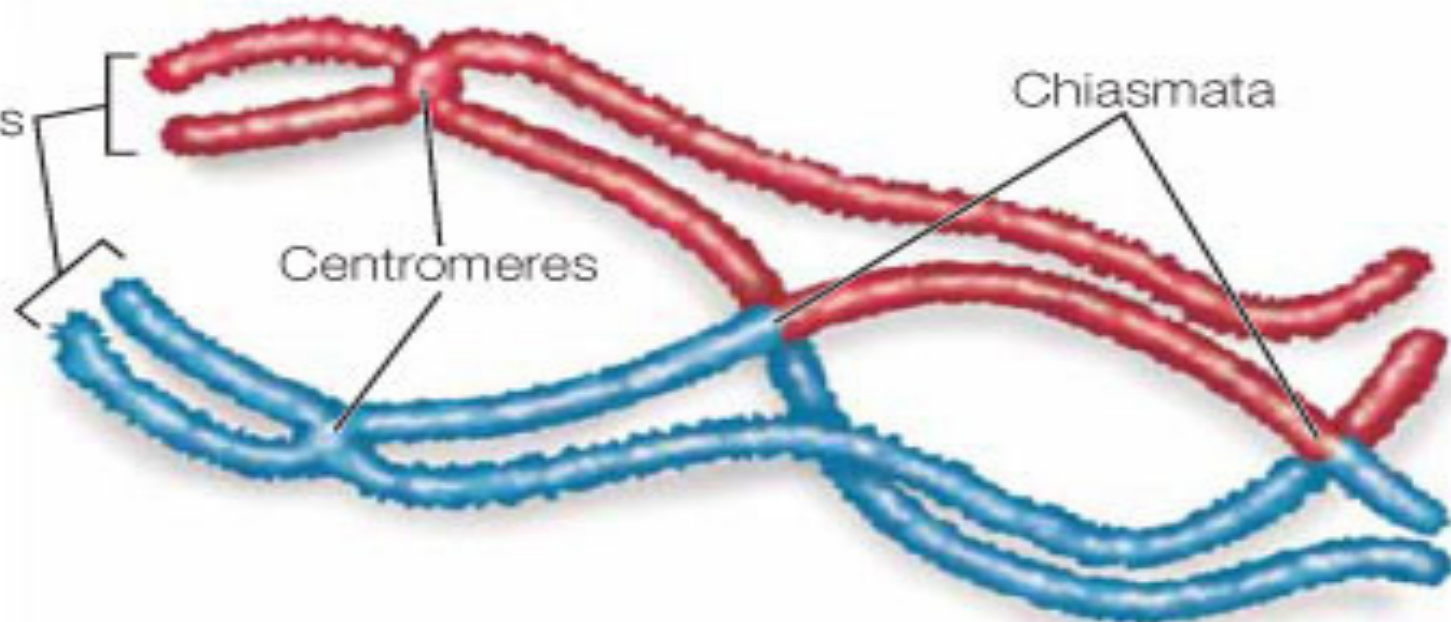
Homologous
chromosomes



Centromeres

Chiasmata

Homologous
chromosomes

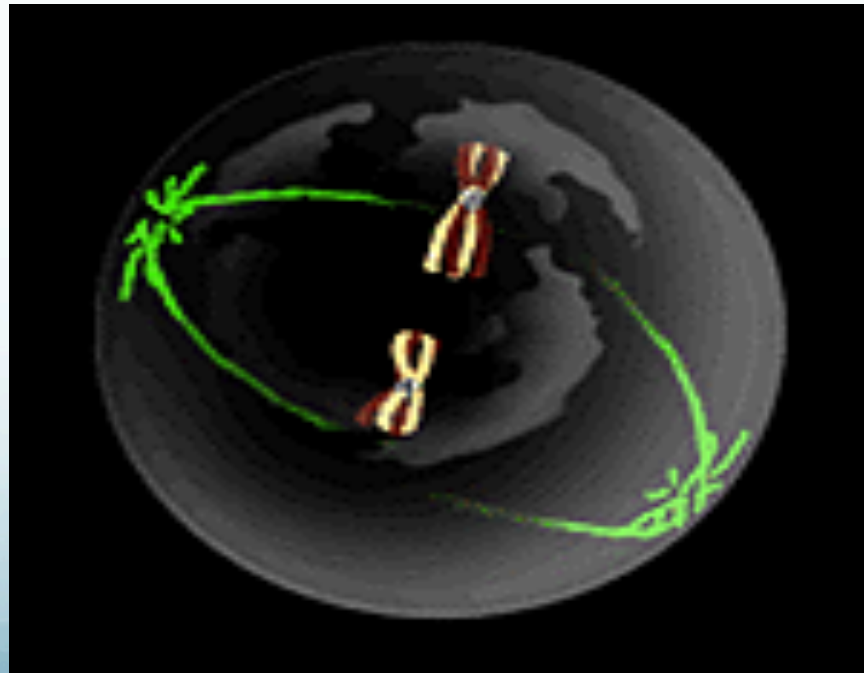


Centromeres

Chiasmata

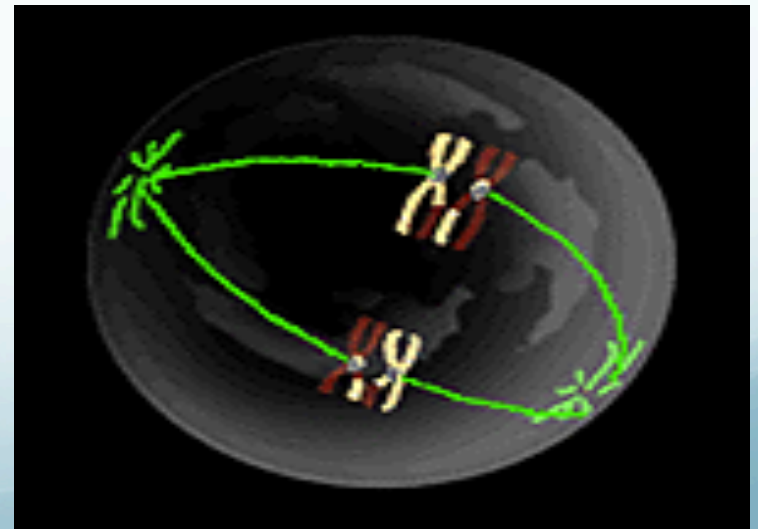
Prometaphase I

- Spindle apparatus formed, and chromosomes attached to spindle fibres by kinetochores.
-



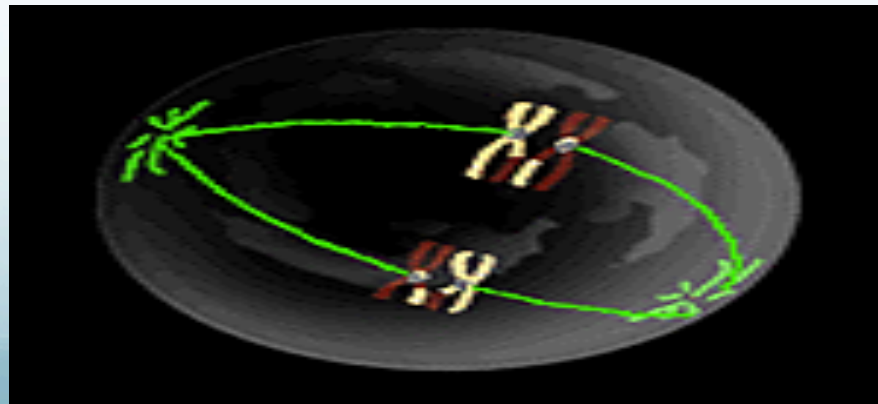
Metaphase I

- Homologous pairs of chromosomes (bivalents) arranged as a double row along the metaphase plate. The arrangement of the paired chromosomes with respect to the poles of the spindle apparatus is random along the metaphase plate.



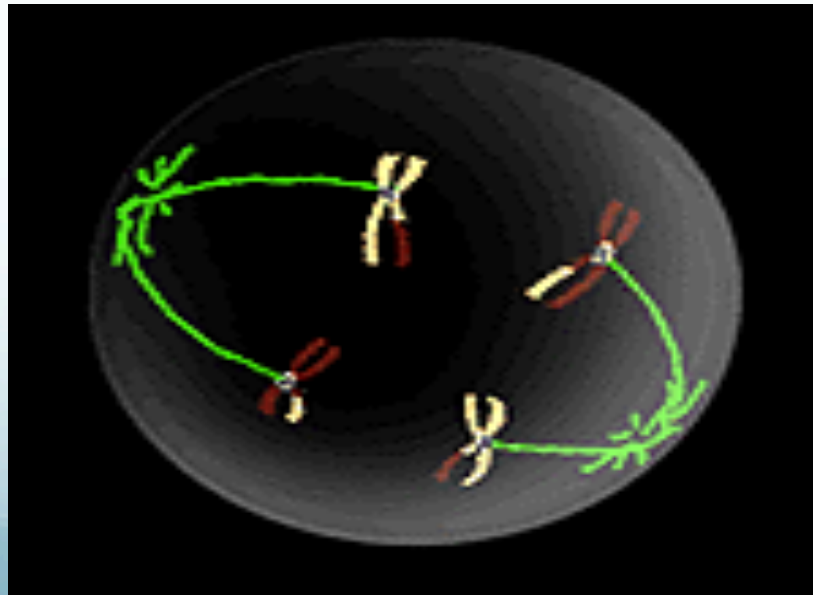
Metaphase I

- (This is a source of genetic variation through random assortment, as the paternal and maternal chromosomes in a homologous pair are similar but not identical. The number of possible arrangements is 2^n , where n is the number of chromosomes in a haploid set.



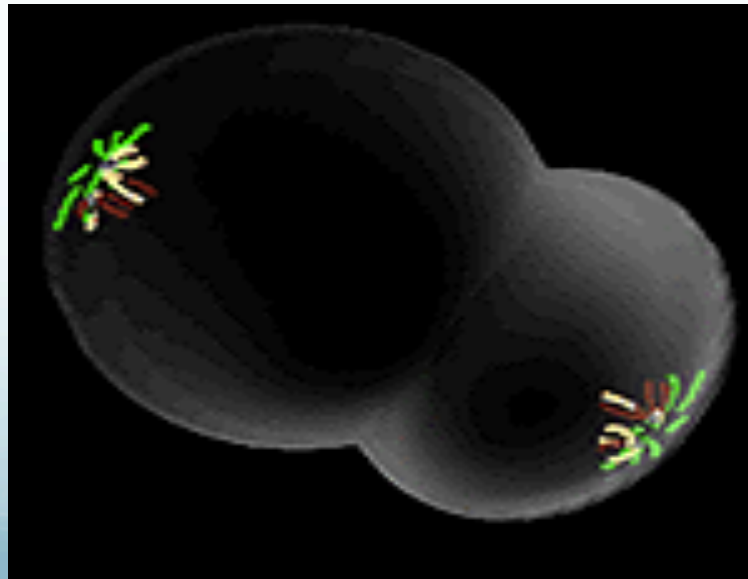
Anaphase I

- The homologous chromosomes in each bivalent are separated and move to the opposite poles of the cell



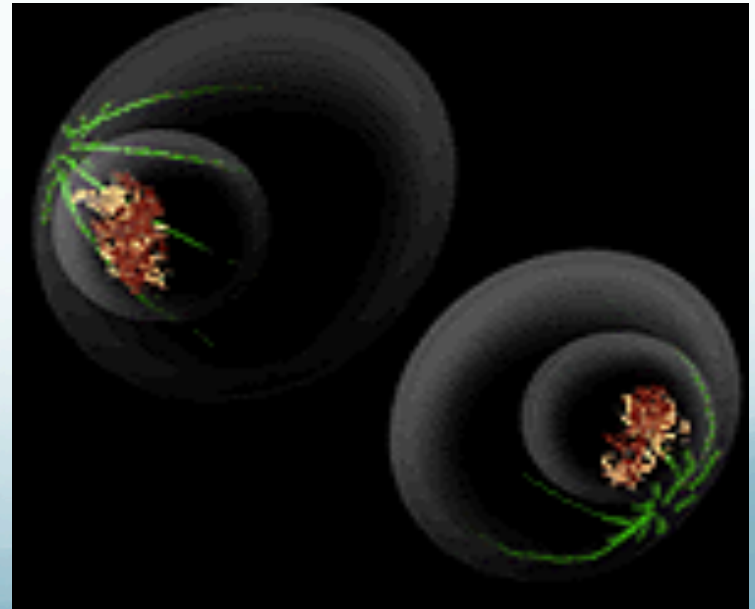
Telophase I

- The chromosomes become diffuse and the nuclear membrane reforms.



Cytokinesis

- The final cellular division to form two new cells, followed by Meiosis II. Meiosis I is a reduction division: the original diploid cell had two copies of each chromosome; the newly formed haploid cells have one copy of each chromosome.



Meiosis II

- Meiosis II separates each chromosome into two chromatids
- The events of Meiosis II are analogous to those of a mitotic division, although the number of chromosomes involved has been halved.

Meiosis

- Meiosis generates genetic diversity through:
- *the exchange of genetic material between homologous chromosomes during Meiosis I
- *the random alignment of maternal and paternal chromosomes in Meiosis I
- *the random alignment of the sister chromatids at Meiosis II

Meiosis I

telophase &
cytokinesis

Meiosis II

prophase II

metaphase II

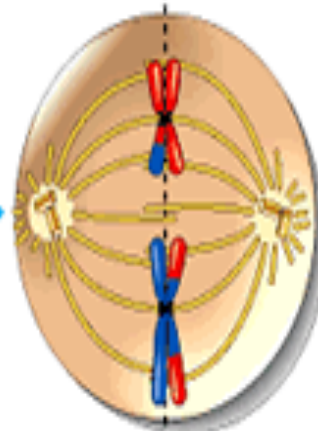
anaphase II

telophase II

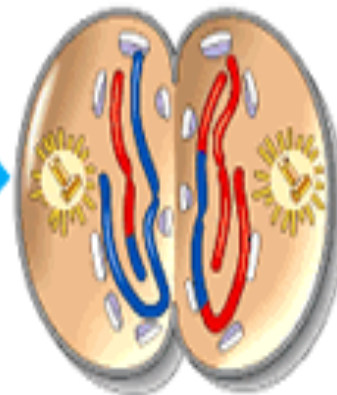
Cleavage
furrow



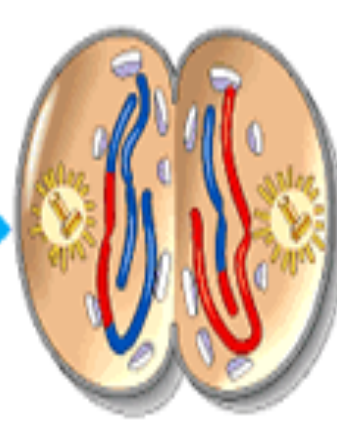
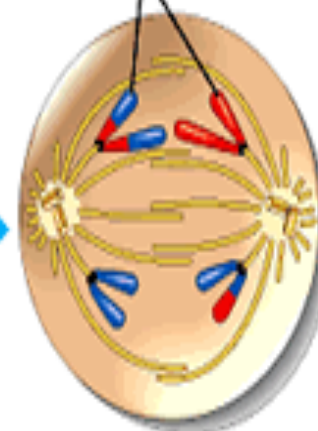
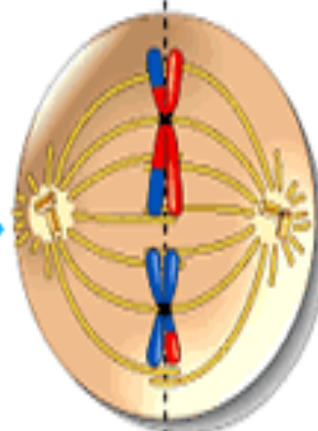
**Two haploid cells
form; chromosomes
are still double**



Sister chromatids
separate



Haploid daughter
cells forming



**During another round of cell division, the sister chromatids finally
separate; four haploid daughter cells result, containing single
chromosomes**

Meiosis I in Females

Prophase I

chromosomes begin to condense

homologous chromosomes pair
crossing over occurs

recombinant chromosomes

Metaphase I

spindle fibers attach to chromosomes
chromosomes line up in center of cell

Anaphase I

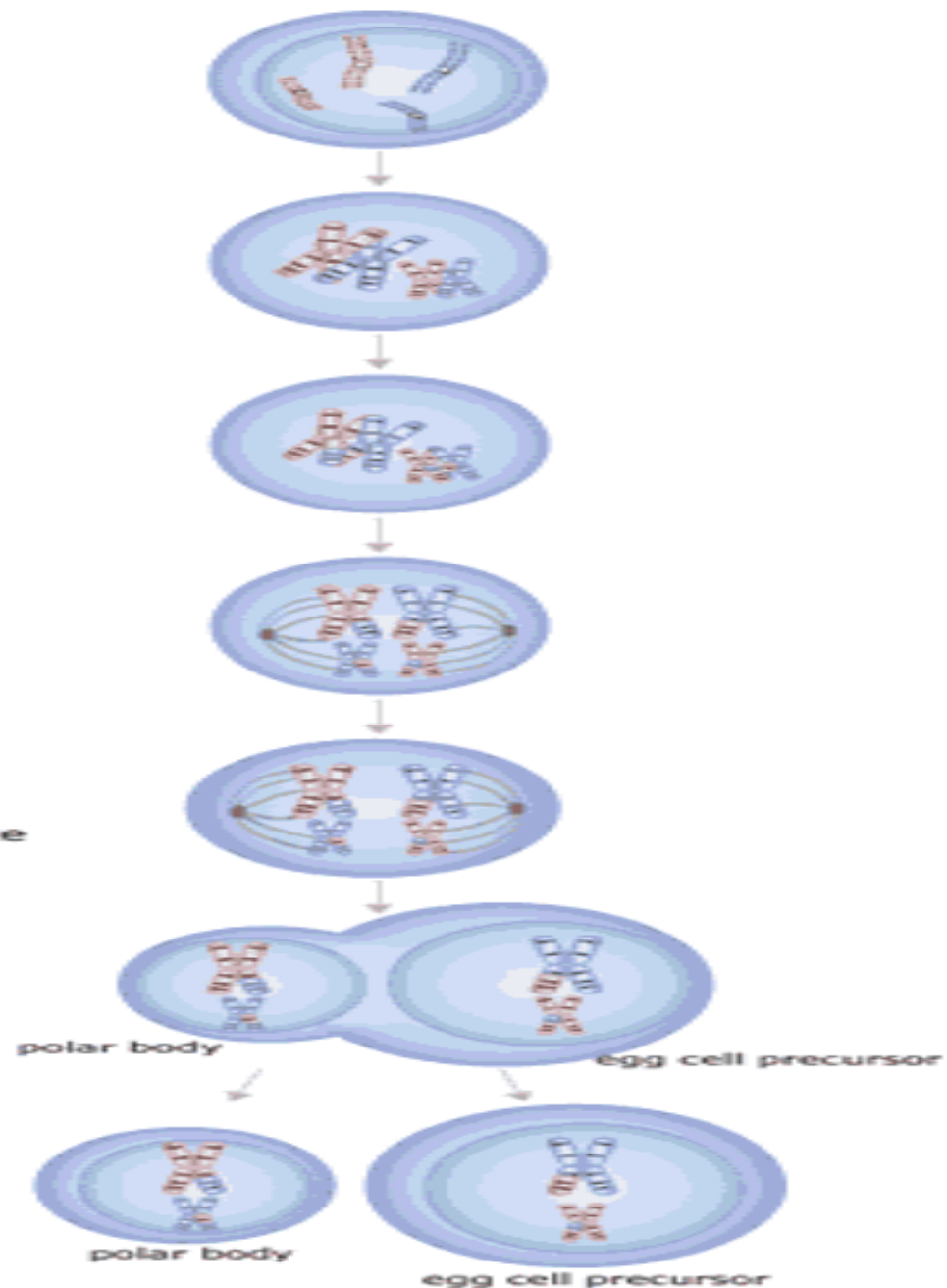
chromosomes start to move to opposite
ends of cell as spindle fibers shorten

Telophase I

chromosomes reach opposite ends
nuclear membrane forms

Cytokinesis

cell division occurs



Meiosis I in Males

Prophase I

chromosomes begin to condense

homologous chromosomes pair
crossing over occurs

recombinant chromosomes

Metaphase I

spindle fibers attach to chromosomes
chromosomes line up in center of cell

Anaphase I

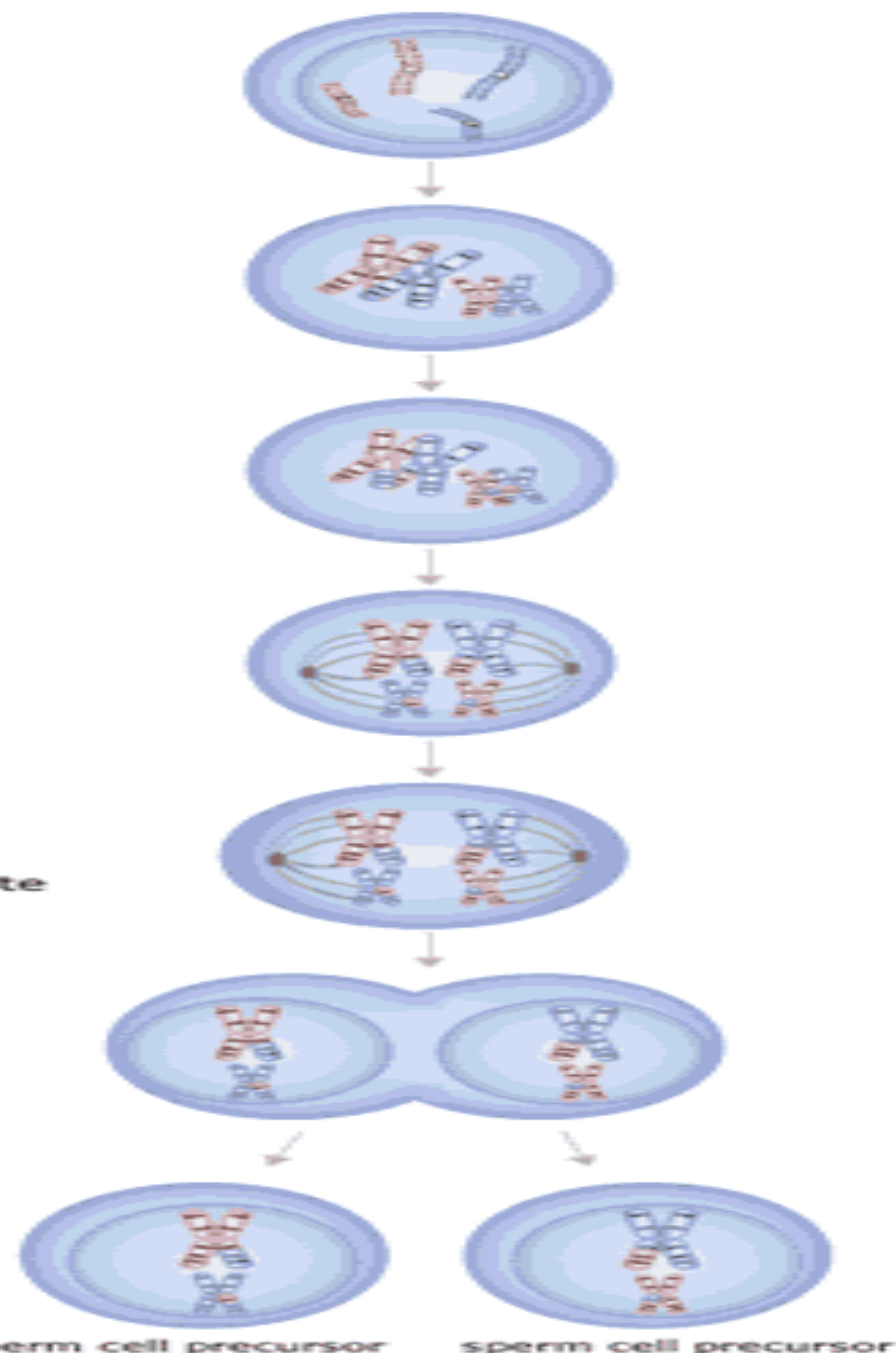
chromosomes start to move to opposite
ends of cell as spindle fibers shorten

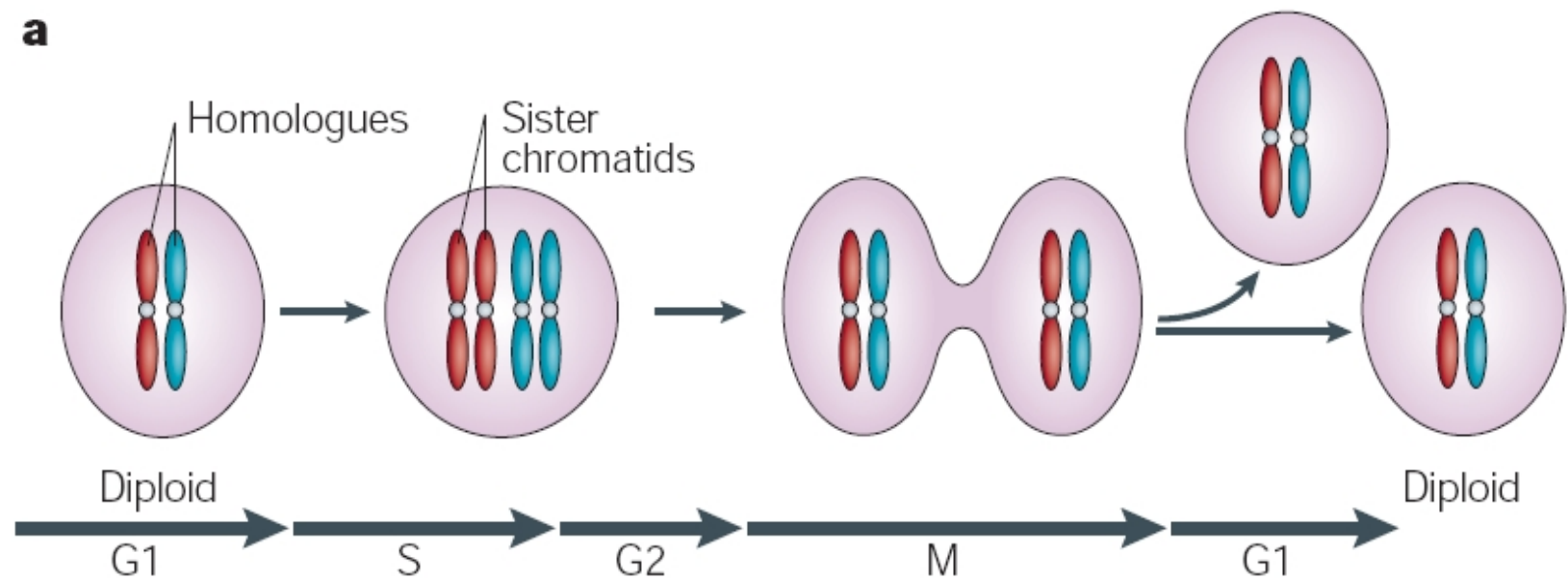
Telophase I

chromosomes reach opposite ends
nuclear membrane forms

Cytokinesis

cell division occurs



a**b**